



Mutations induced by Bleomycin, 4-nitroquinoline-1-oxide, and hydrogen peroxide in the *rpoB* gene of *Escherichia coli*: Perspective on Mutational Hotspots

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ABSTRACT

We report the mutational spectra in a segment of the *E. coli rpoB* gene of bleomycin (BLEO), 4-nitroquinoline-1-oxide (NQO), and hydrogen peroxide (H_2O_2). We compare these spectra with those of other mutagens and repair deficient strains in the same *rpoB* system, and review the key elements determining mutational hotspots and outline the questions that remain unanswered. We consider three tiers of hotspots that derive from 1) the nature of the sequence change at a specific base, 2) the direct nearest neighbors and 3) some aspect of the larger sequence context or the local 3D-structure of segments of DNA. This latter tier can have a profound effect on mutation frequencies, even among sites with identical nearest neighbor sequences. BLEO is dependent on the SOS-induced translesion Pol V for mutagenesis, and has a dramatic hotspot at a single mutational site in *rpoB*. NQO is not dependent on any of the translesion polymerases, in contrast to findings with plasmids treated *in vitro* and transformed into *E. coli*. The *rpoB* system allows one to monitor both G:C -> A:T transitions and G:C -> T:A transversions at the same site in 11 cases, each site having the identical sequence context for each of the two mutations. The combined preference for G:C -> A:T transitions at these sites is 20-fold. Several of the favored sites for hydrogen peroxide mutagenesis are not seen in the spectra of BLEO and NQO mutations, indicating that mutagenesis from reactive oxygen species is not a major cause of BLEO or NQO mutagenesis, but rather specific adducts. The variance in mutation rates at sites with identical nearest neighbors suggests that the local structure of different DNA segments is an important factor in mutational hotspots.

1. Introduction

Investigators have been intrigued by mutagenesis, mutagens, and the mechanism of mutagenic processes since even before the elucidation of the structure of DNA, and of course ever since (e.g. [1–4], see reviews [5–7]). In fact, the history of mutation detection systems in microorganisms spans more than 60 years, and shows the progression of better and more developed systems (e.g. [8–16]). Some of these systems were aimed principally at detecting mutations, and employed a small number of sites (e.g. [10,14,15]), while other were aimed at analyzing the distribution of mutations at a larger number of sites (e.g. [11,13,16]). Now, one is able to sequence whole genomes to analyze mutations that can occur at a very large number of targets [17,18]. Does this technology supplant all of the previous technologies? We

argue here that in fact the ability to sequence whole genomes to analyze mutations complements rather than replaces some of the other systems currently in use. The important facets of single gene systems that we will elaborate below are: 1) relative ease of use and low cost. 2) the ability to generate “fingerprints” that can be used to recognize a specific process. 3) the ability to define hotspots within a fingerprint, which requires tabulating multiple repeat occurrences at particular sites.

Originally, Seymour Benzer's epic studies of the *rII* genes of phage T4 elegantly demonstrated the phenomenon of “hotspots”, mutational sites that have significantly higher rates of mutation than other sites, and in some cases dramatically higher rates [2]. At the time, one did not know the exact sequence change involved in these mutations, but later work from our laboratory [11,19,20] and numerous other groups (e.g. [13,21]) demonstrated that even among mutational sites that exhibited

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the identical base changes, there were still hotspots. Thus, the same base change, or base addition or deletion, or even larger deletions or rearrangements, could occur or be induced at vastly different rates. Clearly, some aspect of the surrounding or perhaps even more distant sequence context is important for mutation rates. This is true for mutations occurring spontaneously in wild-type strains [16,19], or in “mutator” strains with higher mutation rates than normal [13,16,19,22,23], or for mutations induced by mutagens [11,16,20,24]. In fact, for a given mutagen or mutator, one can recognize at least three tiers of preference that lead to the hotspots that we observe. The first tier involves the base or bases that are preferentially affected by a mutagenic agent. For instance, although ultraviolet irradiation (UV) can result in different base substitutions, it preferentially induces C → T (or more strictly G:C → A:T) changes among base substitutions. Thus, C → T represents the first tier of preference. The second tier of preference involves nearest neighbor sequences, as UV preferentially induces mutations at -C- residues that are part of a 5'-T-C-3' or 5'-C-C-3' sequence ([11,20]; for a more complete discussion see below). This is because these sequences are prone to the two types of photodimers induced by UV. Nearest neighbors include the base on each side of the mutated base, so one can say that each altered base pair is part of a triplet. Analyzing favored neighboring bases, and “triplets” is a key part of whole genome sequencing results [17,18,25,26]. The third tier of preference involves some aspect of the DNA sequence that goes beyond the nearest neighbors. Even sites with identical preferential surrounding nearest neighbor sequences still have enormous differences in observed mutations that can approach two orders of magnitude. The second and third tiers of preference result in hotspots for many mutagens and mutagenic processes. In some cases the direct nearest neighbors play a dominant role in defining hotspots (second tier). However, as we shall see below, in many cases it is the third tier that results in the largest differences in mutation frequencies at different sites. The underlying cause of this still remains a mystery, as we have not yet deciphered what aspect of the larger surrounding sequence is responsible. We need a genetic/biophysical “Rosetta Stone” to interpret the “hotspot hieroglyphic”. An important point is that the last and crucially important third tier can only be defined by recording many repeat occurrences at defined sites. This is what single gene systems allow. Whole genome sequencing does not, because one is rarely dealing with repeat occurrences at any site (however, see [18]). In whole genome sequencing there are usually between several hundred thousand to more than a million possible sites.

In this paper we report the mutational spectra in a segment of the *E. coli rpoB* gene of bleomycin (BLEO), a mutagen and potential carcinogen that is also used in chemotherapy [27], and 4-nitroquinoline-1-oxide (NQO), a potent carcinogen [28]. Because these mutagens might act partially or completely via reactive oxygen species, we also determined the spectrum of mutations induced by hydrogen peroxide, an agent that causes oxidative damage. We compare these spectra with those of other mutagens in the same *rpoB* system, and review what we can deduce and still need to determine about the nature of hotspots and the mechanisms of mutagenesis.

2. Materials and methods

***E. coli* strains.** Table 1 lists the strains used. The starting strain is CC107 [15], designated as wild-type, which we have used in multiple mutagenesis studies (e.g. [16,29]). We have not observed any differences in the mutagenesis seen between this strain and in the other wild-type strains (ZK126; BW25113) we have used (e.g. [30]; see below). The starting wild-type strain used in Table 2 is ZK126 [31]. Its derivative, SF2018 [32], is translesion synthesis deficient (TLS⁻), as it is deficient in all three SOS-induced DNA polymerases (Pol II, Pol IV, Pol V). These two strains were a gift from Steven E. Finkel. The Ung, MutM, PolIV, PolV, and MutL-deficient strains used here are from the Keio collection, described in Baba et al. [33], made from the starting

strain BW25113 [34]. Strain EM90 [19] is a derivative of CC107mutS lacking the F' plasmid.

Media. The following media [35] were used. LB (10 g tryptone, 5 g yeast extract, 10 gm NaCl per liter). Minimal A (10.5 g K₂HPO₄, 4.5 g KH₂PO₄, 1 g (NH₄)₂SO₄, 0.5 g sodium citrate·2H₂O per liter), supplemented with 10 ml of 20 % glucose, 1 ml of 1 M MgSO₄, and 0.5 ml of 1% thiamine hydrochloride (vitamin B1), per liter.

Growth conditions, Mutagenesis treatment. Unless otherwise stated, all genetic methods are as described by Miller ([35]; see also [36]). NQO Treatment: Overnight cultures grown in a 37 degree incubator were used to seed “overday” cultures by inoculating 250 µl into separate 5 ml LB cultures that were then incubated in a 37 °C water bath for 3–4 h. After placing on ice for 5 min, the cultures were pooled and spun down, the liquid discarded and the pellets washed in 1 x A minimal media, recentrifuged and resuspended in half the initial LB volume in 1 X A minimal medium. 2 ml samples were aliquoted into clean culture tubes and NQO was added to a final concentration of 50 µg/mL. The NQO was diluted from a freshly prepared stock solution of 5 mg/mL in acetone. Treatment was for 60 min at 37 degrees. The cells were spun down and resuspended into 2.5 ml minimal A medium. These cultures were titered for survival. Outgrowth cultures were seeded with 0.2 ml of this resuspension in 5 ml LB, grown overnight at 37 °C on a rotor at 50 rpm (0.44 g) before plating. BLEO treatment: Cultures were seeded with approximately 500–1,000 cells by inoculating 2 ml cultures with 50 µl of a 10⁻⁴ dilution of a growing culture. The tubes contained 50 µg/mL BLEO. BLEO was prepared by making a stock solution of 2 mg/mL in distilled H₂O, and subsequently diluting into the media. After 18 h incubation at 37 °C on a rotor at 50 rpm 0.5 ml of each culture was subcultured into 4.5 ml LB without BLEO, and these subcultures were grown again overnight and subsequently plated for viable cells and for mutants. H₂O₂ treatment: cells were first grown overnight in supplemented minimal A medium. Cells were aliquoted (2 ml) into separate tubes and 30 % H₂O₂ was added to a final concentration of 5, 10, or 12 µl/mL equaling 12, 24, and 29 mM, respectively. The tubes were placed on a rotor at 37 °C for 20 min at 50 rpm before inoculating 250 µl into 5 ml LB and growing overnight at 50 rpm before plating.

Determination of mutant frequencies. The cells grown as indicated above were plated on LB plates with or without 100 µg/mL rifampicin. The frequencies of Rif^r (rifampicin resistant) mutants were determined as described previously [16,29]. Briefly, mutant frequency (f) was determined as the median frequency from a set of cultures (N = number of cultures). 95 % confidence limits were determined according to Dixon and Massey [37].

Chromosomal DNA isolation and sequencing. Chromosomal DNA was isolated from overnight cultures of each mutant after single colony purification by streaking. The PCR tests were carried out as colony PCR reactions. The *rpoB* gene was PCR-amplified from genomic DNA using Taq polymerase (Biorad) and sets of primers which allowed us to sequence directly from the PCR product. The sequences of the primers used for amplifying the *rpoB*-encoding gene were as follows: 5'-CGTCGTATCCGTTCCGTTGG-3' and 5'-TTCACCCGGATAACATCTC GTC-3', for the main set of mutations (cluster II). The unpurified PCR product was outsourced to Laragen (Culver City, CA) for purification with exoSap (Affymetrix, USA) and sequencing. The following primer for the main cluster II was used for sequencing: 5' -CGTGTAGAGCGT GCGGTGAAA-3'. Virtually all of the mutations resulting in Rif^r mutants were detected with this primer. Mutant sequences were compared to the *rpoB* wild-type sequence using ApE (M. Wayne Davis) software.

Chemicals. Rifampicin, 2-aminopurine (2AP), NQO, EMS, and trimethoprim (TMP) were purchased from Sigma (St. Louis, MO). Bleomycin was purchased from EMD Millipore Corp. (Billerica, MA).

3. Results

Bleomycin (BLEO). Table 2 shows the levels of BLEO induced

Table 1
Bacterial strains described in this report.

Strain	Genotype	Key Phenotype	Reference
CC107	<i>ara</i> Δ (<i>gpt-lac5</i>) <i>thi</i> /F ⁺ 128 <i>lacIZ</i> <i>proA</i> ⁺ <i>B</i> ⁺ (carries a +1 frameshift in <i>I</i> portion of <i>I-Z</i> fusion)	–	[35]
CC107 <i>mutS</i>	CC107 <i>mutS</i> ::mini-Tn10 <i>cat</i>	MMR [–]	[29]
ZK126	W3310 Δ <i>lacU169</i> <i>tna-2</i>	–	[31]
SF2018	ZK126 <i>polB</i> ::Spc ^r <i>dinB</i> ::Kan ^r <i>umuDC</i> ::Cam ^r	TLS [–]	[32]
BW25113	<i>lacI</i> ^r <i>rrnB</i> _{T14} Δ <i>lacZ</i> _{WJ16} <i>hsdR514</i> Δ <i>araBAD</i> _{AH33} Δ <i>rhaBAD</i> _{LD78}	–	[34]
BW25113 <i>ung</i>	BW25113 <i>ung</i> ::Kan ^r	Ung [–]	[33]
BW25113 <i>mutM</i>	BW25113 <i>mutM</i> ::Kan ^r	MutM [–]	[33]
BW25113 <i>dinB</i>	BW25113 <i>dinB</i> ::Kan ^r	Pol IV [–]	[33]
BW25113 <i>umuDC</i>	BW25113 <i>umuDC</i> ::Kan ^r	Pol V [–]	[33]
BW25113 <i>mutL</i>	BW25113 <i>mutL</i> ::Kan ^r	MutL [–]	[33]
EM90	<i>ara</i> Δ (<i>gpt-lac5</i>) <i>mutS</i> ::mini-Tn10 <i>thi</i>	MutS [–]	[19]

mutations in several backgrounds. BLEO induces mutations in the CC107 wild-type strain 17-fold over the spontaneous background. (The induction over background in two other wild-type strains is higher, although the absolute level of mutagenesis in all three strains varies within only a 2.4-fold range). BLEO-induced mutagenesis is dependent on the SOS translesion polymerase Pol V, but not Pol IV (Table 2). BLEO does not appear to induce mutations above the higher background in a strain lacking mismatch repair (MMR[–]; *mutS*; data not shown). Fig. 1 displays the spectrum of BLEO induced mutations in the *rpoB* gene that result in Rif^r mutants. Here we see that there is one prominent hotspot at 1576 consisting of 146 of the 190 sequenced mutations. Of these, 126 are G:C -> A:T transitions, and represent virtually all (126 of 138) of the G:C -> A:T transitions detected. Although the 20 transversions detected at 1576, as well as the 24 transversions detected at the neighboring 1577 could be induced by BLEO, they could also be attributed to the SOS polymerase errors across from undamaged base pairs. This is because the transversions at 1576 are a known hotspot for Pol IV [19], and those at 1577 are a hotspot for some homologs of Pol V [38]. Therefore, BLEO possibly directly targets only a single mutation at a single site in *rpoB*. The hotspot at 1576 may be a result of the nearest neighbor sequence (second tier hotspot) and/or the extended sequence context (third tier hotspot), as considered in the Discussion.

4-nitroquinoline-1-oxide (NQO). Table 3 displays the data for 4NQO in *rpoB*. NQO is a more powerful mutagen than BLEO, and induces mutations more than 200 fold over background. Unlike BLEO, NQO is not dependent on one or more of the SOS polymerases for mutagenesis, as can be seen by the lack of significant reduction in mutation frequency in a strain lacking all three SOS-induced polymerases compared with the control strain. (We also found no dependence on Pol IV or Pol V using strains lacking either one of these polymerases in both the BW25113 and ZDK strain backgrounds; data not shown). The lack of dependence on the SOS polymerases for NQO

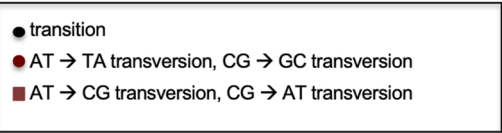
mutagenesis contradicts previous results [39,40], which showed a dependence on SOS functions, and particularly Pol V for mutagenesis. These experiments were done with plasmid DNA that was altered *in vitro* and then transformed into UV irradiated cells. Mutagenesis was monitored by scoring tetracycline resistance emanating from a plasmid borne *tet* gene. It is possible that the DNA replication apparatus operating on chromosomal DNA treats NQO adducts differently than the replication system acting on a pBR322-based plasmid. Fig. 2 shows the distribution of mutations in *rpoB*. Here, virtually all the mutations are G:C -> A:T transitions, and they are essentially evenly distributed amongst 5 of the 14 possible G:C -> A:T transition sites, with several mutations occurring at a sixth site. Several adducts involving guanine residues have been implicated in NQO mutagenesis, including the N2 position, deemed the most important for mutagenesis, and also the generation of 8-oxodGuanine adducts that might be expected to lead to G:C -> T:A transversions ([39,40], and references therein). Such transversions have certainly been reported as a minor component of NQO mutagenesis *in vivo*, (e.g. [11]), and NQO reverts CC104 to Lac⁺ via the G:C -> T:A change under the conditions employed here (data not shown). However, these transversions are not prominent in the *rpoB* system in Fig. 2. Also, oxidative damage to cytidine residues that can lead to deamination of cytidine to uracil has been suggested for some mutagens [41]. Therefore, we looked at NQO mutagenesis in a strain (*mutM*) lacking the MutM protein, a glycosylase that removes 8-oxodGuanine [42], as well as in a strain (*ung*) lacking the Ung protein that removes uracil from DNA [43]. In addition, we looked at NQO mutagenesis in a strain lacking mismatch repair (*mutS*). Fig. 2 shows the spectrum of NQO mutations in the same CC107 background as used for BLEO (Fig. 1), and spontaneous mutations (see below). Fig. 3 shows the NQO spectrum in a series of backgrounds, compared with a larger set of NQO mutations as compiled from three different wild-type backgrounds. Out of 195 mutations in this larger sample, 182 are G:C ->

Table 2
rpoB mutant frequencies (f) in various knockout strains of *E. coli* with Bleomycin.

Strain	Treatment	Concentrations (μ g/mL)	Number of Replicates (N)	Frequency f (10^{-8}) ^a in <i>rpoB</i>
WT, CC107	None	–	41	3.7 (3.1, 5.6)
	BLEO	0.25	18	64.6 (29.9, 119)
	BLEO	0.33	29	128.0 (72.8, 156)
	BLEO	0.40	26	118.0 (79.0, 163)
WT, ZK126	None	–	26	1.1 (0.3, 2.3)
	BLEO	0.25	5	136.0 (89.8, 166)
SF2018 (TLS [–])	None	–	26	0.7 (0, 1.7)
	BLEO	0.25	5	5.1 (2.9, 7.2)
WT, BW25113	None	–	9	2.5 (0.8, 2.9)
	BLEO	0.25	15	56.4 (41.9, 104)
BW25113 <i>dinB</i> Pol IV [–]	None	–	9	1.5 (0, 12.0)
	BLEO	0.25	15	45.8 (27.0, 73.1)
BW25113 <i>umuC</i> Pol V [–]	None	–	9	3.0 (0, 5.7)
	BLEO	0.25	15	2.6 (0.7, 6.5)

^a Values in parenthesis are 95 % confidence limits.

Bleomycin (BLEO)



192 mutations

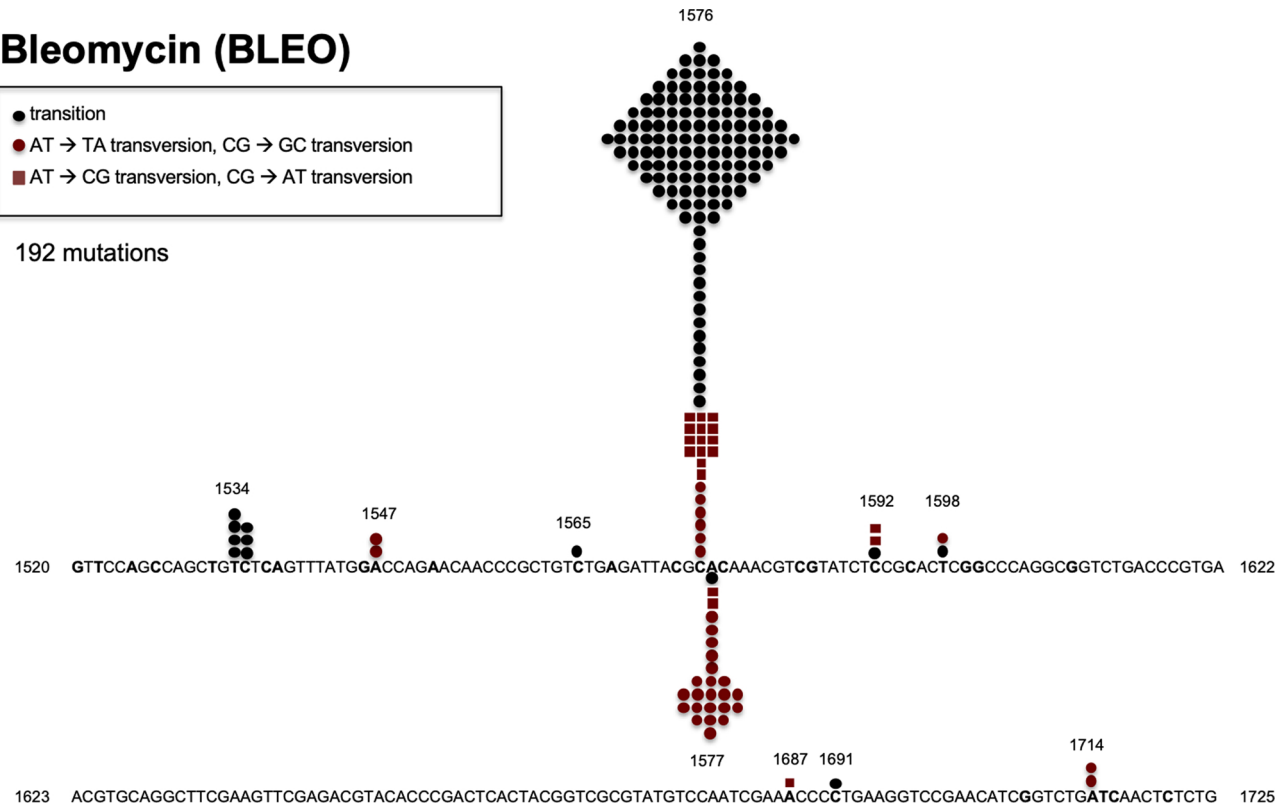


Fig. 1. Distribution of bleomycin-induced mutations in *E. coli* *rpoB* that lead to rifampicin resistance. Strain CC107 was used.

A:T changes, 10 are G:C -> T:A changes, 2 are G:C -> C:G changes, and 1 is an A:T -> T:A change. Work with plasmid DNA treated with NQO (referred to above) resulted in an approximately equal number of G:C -> T:A transversions and G:C -> A:T transitions for NQO [39,40], a result not seen here. Also, complete genomic sequencing of 3994 mutations in *Aspergillus nidulans* [44] revealed that changes at G:C base pairs were 19-fold more frequent than changes at A:T base pairs, with G:C -> A:T changes predominating by only a 2:1 ratio over G:C -> T:A changes, and a 4:1 ratio over G:C -> C:G changes. However, a positive feature of the well characterized *rpoB*/Rif^r system is the set of 11 defined sites that can generate a Rif^r mutant by either a G:C -> A:T or a G:C -> T:A change at the same base pair [19,45]. Here, we can compare the two possible events within the same exact sequence context. At 7 of these positions we find NQO-induced mutations leading to Rif^r. The ratio of G:C -> A:T changes to G:C -> T:A changes at these sites is as follows: position 1535, 8:0, position 1546, 44:1, position

1565, 40:1, position 1576, 8:5, position 1586, 24:1, position 1592, 18:1, position 1691, 36:0. The total of 176:9 for the 7 sites shows the high preference for the G:C -> A:T transition. There are also three sites at which we found mutations leading to Rif^r that can mutate either via a G:C -> A:T or a G:C -> C:G change. Here the numbers for the respective changes are: 1576, 8:1, 1592, 18:1, and 1691, 36:0 (total = 52:2). Moreover, the few transversions at 1576 are possibly due to the action of the SOS polymerase Pol IV, as this is a favored change when Pol IV is over-expressed [19] in the absence of exogenous DNA damage. Even though Pol IV is not required for NQO mutagenesis here, it is probably partially induced. The differences between these results very probably reflect the difference in treating cells *in vitro* on plasmids and then transforming *versus* treating cells directly and monitoring chromosomal mutations, and also differences between *E. coli* and *A. nidulans*.

From Fig. 3, the spectrum appears essentially unchanged in almost

Table 3
rpoB mutant frequencies (f) in various knockout strains of *E. coli* with 4NQO.

Strain	Treatment	Concentrations (ug/mL)	Number of Replicates (N)	Frequency f (10 ⁻⁸) ^a in <i>rpoB</i>
WT, BW25113	None	–	83	2.5 (0.8, 2.9)
	4NQO	50	111	570 (490, 641)
BW2553ung-	None	–	129	16.5 (10.5, 24.2)
	4NQO	50	137	764 (531, 840)
BW25113mutM	None	–	39	4.5 (0.6, 5.2)
	4NQO	50	118	1720 (1330, 1850)
WT, CC107	None	–	17	2.2 (1.1, 3.2)
	4NQO	50	145	1360 (1020, 2140)
CC107mutS	None	–	12	909 (636, 1690)
	4NQO	50	140	1610 (1280, 2120)
WT, ZK126	None	–	18	8.0 (2.3, 22.2)
	4NQO	50	29	1170 (1030, 1330)
SF2018 (TLS ^{-/-})	None	–	18	2.7 (1.6, 6.3)
	4NQO	50	29	1470 (1160, 1760)

^a Values in parenthesis are 95 % confidence limits.

4-Nitroquinoline 1-oxide (4NQO)

● transition

● AT → TA transversion, CG → GC transversion

■ AT → CG transversion, CG → AT transversion

104 mutations

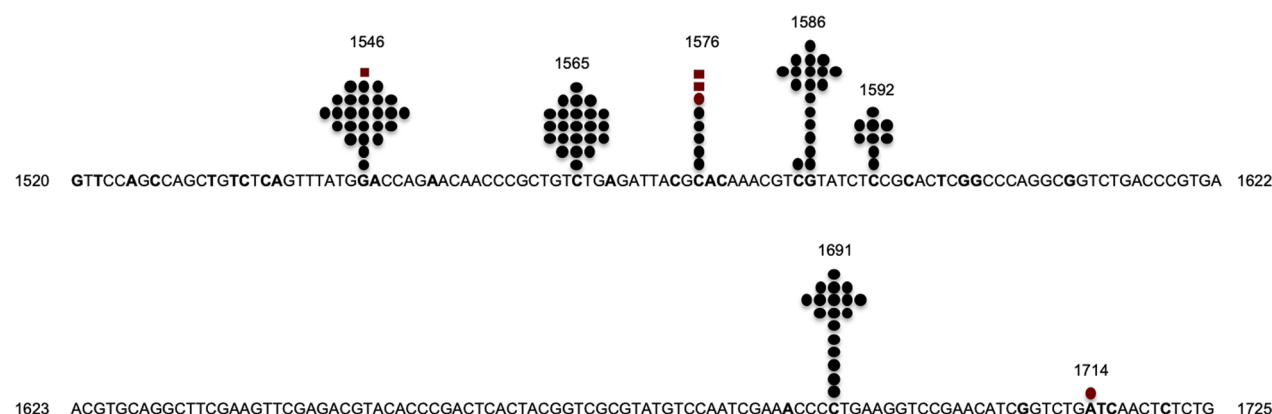


Fig. 2. Distribution of 104 NQO-induced mutations in *E. coli* *rpoB* strain CC107.

all of the repair deficient backgrounds, indicating that the Ung and MutM glycosidases do not play a major role in avoiding mutagenesis by NQO. As foreshadowed by the frequency results in Table 3, the spectrum in a strain lacking all three SOS polymerases is not dramatically different from that in the wild-type. The one partial exception is the spectrum in a mismatch repair (MMR, *mutS*)-deficient strain, which is a mixture of the spectra of both NQO and MMR⁻ with one additional peak of mutations at 1577 (A:T → G:C). This can be attributed to replication errors resulting from the induction of Pol IV,V that are normally efficiently repaired by the MMR system.

Hydrogen Peroxide (H₂O₂). As seen in Table 4, H₂O₂ induces

mutations in *rpoB* only 4–8 fold over background, so one would expect to see an intrusion of the spontaneous background in its mutation spectrum. This is indeed the case (Fig. 4), as the hotspots at 1547 (A:T → G:C), and 1714 (A:T → CG) are seen here, with the former a good indicator of the maximal level of spontaneous mutations. We can see the mutations that are prominent here but not present in the spontaneous spectrum (reproduced in a later figure). Foremost among these are the G:C → T:A transversions at 1546 and 1565. The value of this is that their absence in the two spectra treated above, namely BLEO and NQO, indicates that oxidative damage caused by reactive oxygen species occurring after H₂O₂ treatment is not responsible for the mutations

4-Nitroquinoline 1-oxide

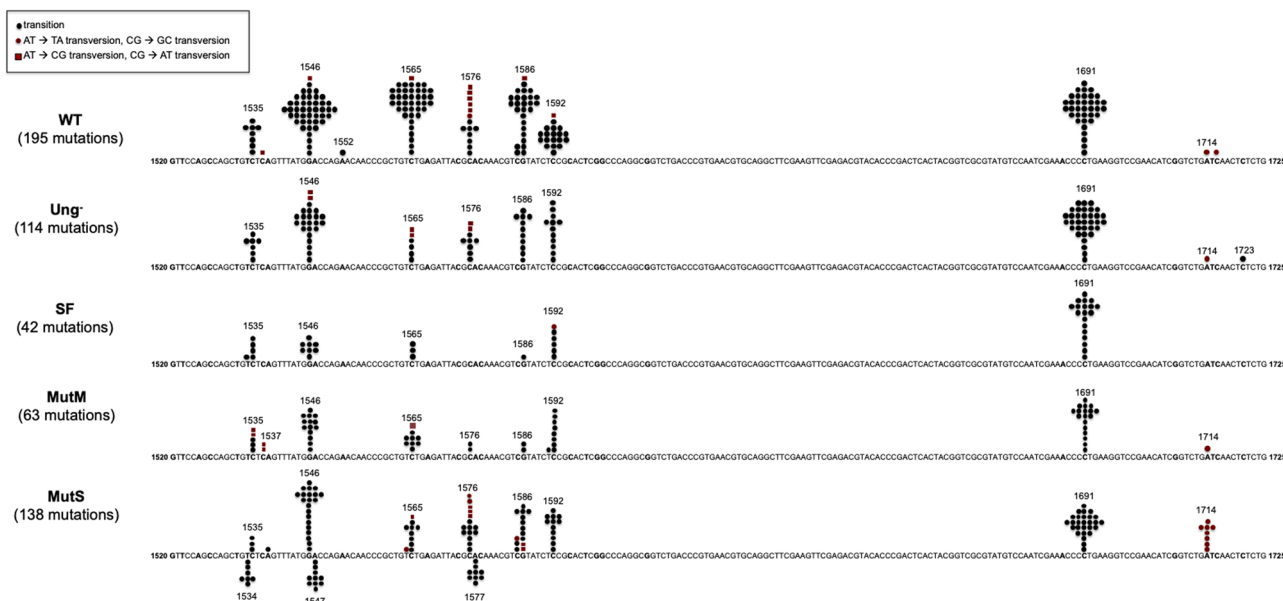


Fig. 3. Distribution of NQO mutations in wild-type and different repair deficient backgrounds. The top panel combines results from three different wild-type backgrounds (CC107, ZK126, and BW25113, see Table 1). The remaining panels show NQO-induced mutations in repair deficient derivatives of BW25113 from the KEIO collection [33].

Table 4
BW25113*mutM*. *rpoB* mutant frequencies (f) in WT and Ung[−] strains of *E. coli* with Hydrogen Peroxide.

Strain	Treatment	Concentrations (mM)	Number of Replicates (N)	Frequency f (10 ^{−8}) ^a in <i>rpoB</i>
WT, CC107	None	–	41	3.7 (3.1, 5.6)
	H ₂ O ₂	12	17	22.9 (22.3, 32.6)
	H ₂ O ₂	24	81	26.9 (24.4, 31.1)
	H ₂ O ₂	29	59	31.2 (28.4, 33.8)
BW25113 <i>ung</i> [−]	None	–	14	21.0 (12.5, 130)
	H ₂ O ₂	12	23	54.8 (29.5, 128)
	H ₂ O ₂	24	23	32.7 (28.9, 42.9)

^a Values in parenthesis are 95 % confidence limits.

Hydrogen Peroxide (H₂O₂)

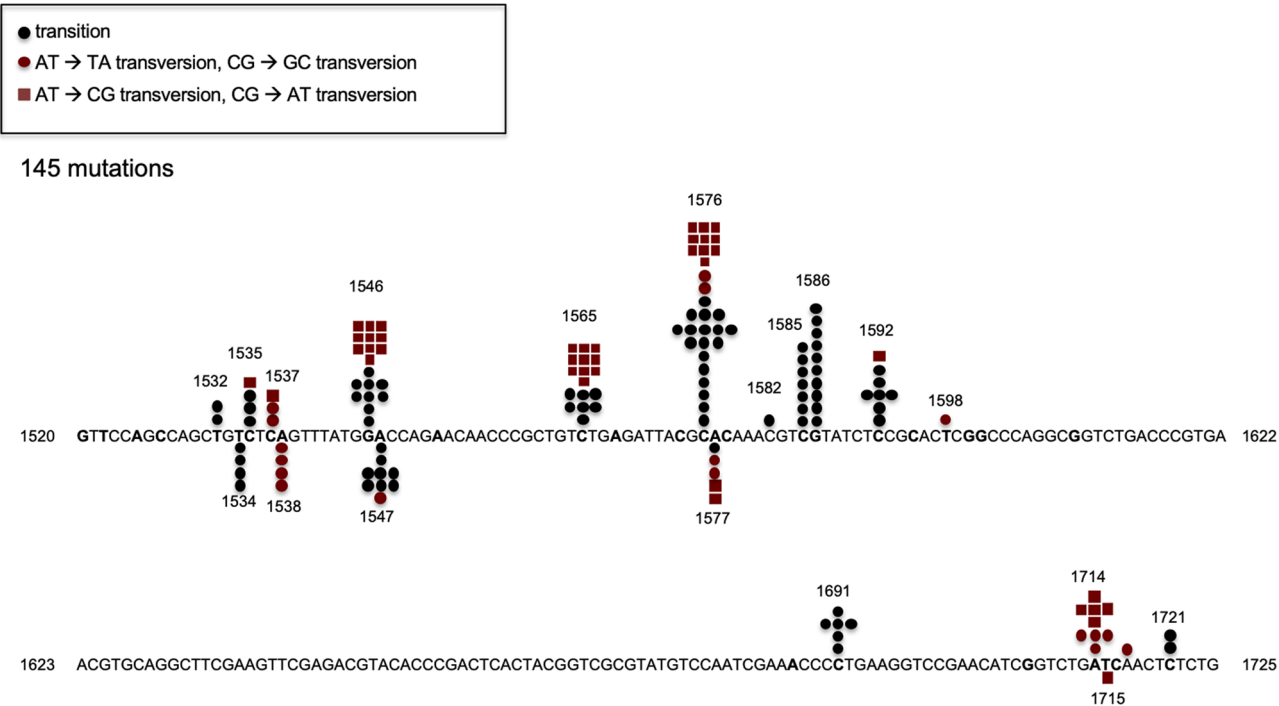


Fig. 4. Distribution of hydrogen peroxide induced mutations in *E. coli rpoB* strain CC107.

observed after BLEO and NQO treatment. Instead, the mutations are very probably caused by adducts or other direct damage by these agents.

Additional results. We sequenced 114 mutations in the Ung[−] background (lacking uracil DNA glycosidase), 189 EMS-induced mutations to supplement the 40 previously sequenced EMS mutations [16], as well as additional mutations in the mismatch repair deficient MMR- (*mutL*) background. We will consider these results in the extended Discussion section below.

4. Discussion: hotspot analysis

One of the fascinating aspects of mutational hotspots is that despite the enormous amount of information gathered, there are still important unanswered questions. Although the field has uncovered much about the sequence biases of different mutagens and mutational processes, we still do not understand the basis of preferences among mutational sites that have the same nearest neighbor sequences. Simply looking at the linear sequence of the DNA directly surrounding mutational sites is not sufficient. Clearly, the larger sequence context plays a dominant role, in some cases perhaps by imparting a local three-dimensional structure to different small segments of DNA. The following figures line up the mutational spectra we have obtained for different mutagens and

mutators. The 200 base pair sequence shown encompasses the region of *rpoB* that is altered by most of the mutations leading to Rif^r. Some of these spectra are from charts and have never been displayed in a figure with an accurate distance metric. We first group the spectra resulting from DNA damaging agents, and then consider those from base analogs, and finally spontaneous mutations from both the wild-type and several repair-deficient backgrounds. The accompanying tables list the set of sites with their neighboring base pairs that have been identified to yield mutations we can score. In the case of *E. coli rpoB*, only mutations that lead to viable Rif^r mutants are monitored. Those that significantly impair RNA polymerase function are not scored. However, we and others have characterized 92 base substitution mutations that result in Rif^r colonies ([16,19,45]; see also Table 5). Let us examine the preferences revealed by numerous studies.

UV irradiation. As a starting example, let us examine the results of UV in the *rpoB* gene. Fig. 5 (Top Panel) shows the position of 40 mutations in the *rpoB* gene leading to Rif^r [16]. Here, we can see a single prominent hotspot (position 1721) among the 14 sites at which a C to T change can be scored, despite the fact that in 10 cases the C is preceded by a T or a C (Table 5). If we consider the neighboring base on each side, meaning the triplet of which the C is the central base, then the hotspot at position 1721 is part of a 5'-TCT-3' sequence. However, the sites at 1535 and 1565 are also part of a TCT triplet (Table 5), yet

Table 5
Triplets 5' – mutated base – 3'.

C → T/G → A	<i>rpoB</i> →	T → C/A → G	<i>rpoB</i>
TCT/AGA	1535 ^a , 1565, 1721	TAT/ATA	
TCC/GGA	1546, 1592	TAC/GTA	
TCG/CGA	1585	TAA/TTA	
TCA/TGA		TAG/CTA	
CCT/AGG	1691	CAT/ATG	
CCC/GGG		CAC/GTG	1577
CCG/CGG	1600, 1609, 1708	CAA/TTG	
CCA/TGG		CAG/CTG	443, 1532, 1538
GCT/AGC		AAT/ATT	
GCC/GGC	1601	AAC/GTT	
GCA/TGC	1576	AAA/TTT	
GCG/CGC		AAG/CTT	
ACT/AGT		GAT/ATC	1714, 1715
ACC/GGT	1520	GAC/GTC	1534, 1547
ACA/TGT		GAA/TTT	
ACG/CGT	1586	GAG/CTC	1598

^a Position of mutated base in *rpoB*.

neither of these are hotspots. Clearly, and again, some additional aspect of the sequence or structure of the DNA is crucial for determining which of the TCT triplets, in this case, are hotspots for UV. If one just considers the 5'-neighbor, then there are three additional 5'-T-C-3' sites. A more in depth look at this phenomenon can be seen in data we presented previously in the *lacI* gene of *E. coli* [20]. Here, more than 1000 UV-induced nonsense mutations were collected and assigned to one of numerous characterized sites. Because the sequences that are converted to either TAG or TAA by a C to T change are part of a NCA triplet, where N is any base, and the C is converted to a T, then we can examine a larger set of each of the four different triplets (TCA, CCA, ACA, GCA). Table 6 displays the results for 662 UV-induced G:C -> A:T mutations. It can be seen how important the 5'-C or T is to the C being mutated (second tier; compare the right and left columns). In addition, it is clear that among these sequences there is an additional factor that determines which of these are true hotspots. Thus, the number of occurrences at a TCA triplet ranges from 7 to 145. Moreover, sites O13 and O24 have identical ATCAA sequences, with the middle C being mutated, but have 7 and 75 occurrences, respectively (third tier)!

Cisplatin (CPT). The spectrum of CPT in *rpoB* ([16]; Fig. 5, Middle Panel) shows one dramatic hotspot, at 1538, for the A:T -> T:A substitution, reminiscent of that seen in Fig. 1 for BLEO for the G:C -> A:T change. This hotspot is part of a CAG (CTG on the opposite strand) sequence, the middle A being affected. However, the *rpoB* gene has two

other CAG/CTG sites that can mutate via the A:T -> T:A change, one at the very nearby 1532, and the other in the small region around 443 that is not part of this figure, but was also monitored when the CPT data were compiled [16]. Again, this is a stunning example of identical nearest neighbors surrounding sites with dramatically different mutation rates for a particular mutagen (third tier).

Ethylmethane sulfonate (EMS). We previously sequenced 40 mutations in *rpoB* generated by EMS [16]. Here, we have sequenced an additional 189 mutations and displayed the spectrum of the total of 229 mutations in Fig. 5 (Bottom Panel). Since EMS generates mutations mainly via the alkylation of the O⁶ position of guanine, the vast majority of its mutations are G:C -> A:T (e.g. [11,14]). A number of the available sites are mutated in this sample, but clearly there are hot and cold spots. Table 7 compares the data for EMS from this study with a previous study in *lacI* by Pienkowski, Glickman, Zielenska and coworkers [13]. It can be seen that, again, sites with identical surrounding sequences can display different mutation rates. Considering both systems, the three examples of the GGA triplet are well mutated, but the AGA triplet is poorly mutated in *rpoB* (three examples) and well mutated in *lacI* (two examples). The triplets CGT, CGC, and CGG show large variations among the several examples of each. With EMS, as with UV, all three tiers are involved (Table 7).

Bleomycin and 4-nitroquinoline-1-oxide. The results in this paper show a different pattern for each of these mutagens. BLEO induces mutations directly at only one site in *rpoB*, which is the sole site among those available that can mutate from C -> T as part of a GCA triplet (position 1576). Moreover, Povirk and coworkers [46], studied phage lambda DNA transfected into UV-irradiated *E. coli* after treatment *in vitro* with BLEO. They found that the mutations in the *CI* gene they could monitor were virtually all at 5'pyrimidine-G-C-3' sequences, with the C being mutated. This sequence corresponds to the favored strand break and apyrimidinic lesions induced by BLEO [47,48]. Despite the low numbers of mutations sequenced (21 mutants with mutated G:C base pairs), 19 of the 21 were part of a Pyr-G-C sequence (the C being mutated [46];). In the *rpoB* work reported here, the virtually single mutated G:C site is also part of a Pyr-G-C sequence, the only one in *rpoB* among the 14 available site to mutate via the C -> T change. Clearly, it will take an analysis of hotspots for BLEO in other system that allow recording multiple repeat occurrences to determine whether other hotspots conform to "Povirk's Rule" or not.

NQO induced G:C -> A:T changes predominate, as seen in Fig. 1, and for an expanded set of NQO-induced mutations shown at the top of Fig. 3. Of the possible 14 sites for this bp change, 6 are well represented and a few more are represented by only a few occurrences. Thus, the

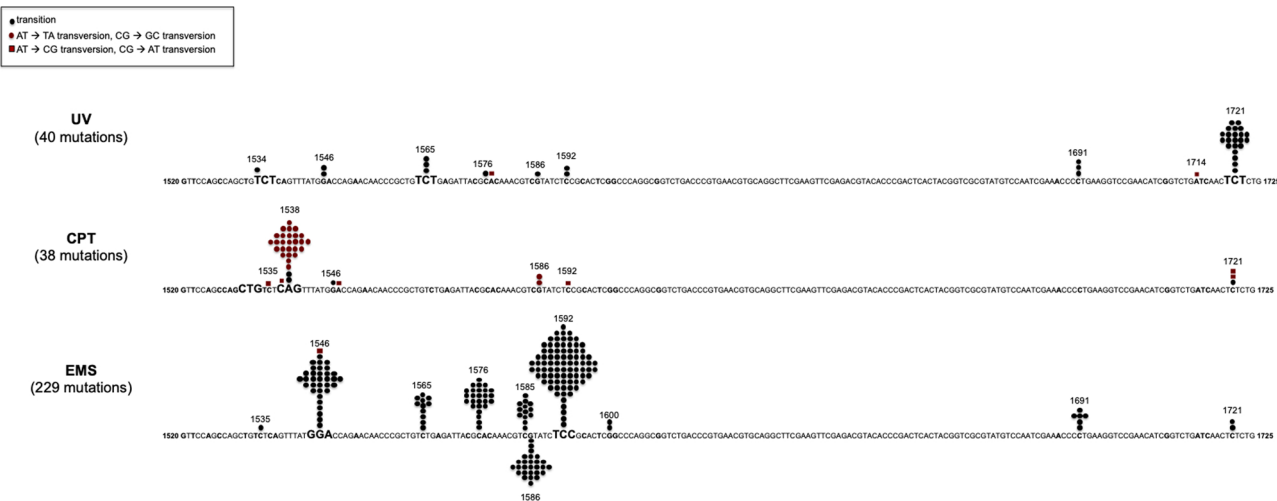


Fig. 5. Distribution of UV, Cisplatin (CPT), and EMS-induced mutations in *E. coli rpoB* in strain CC107. Data are replotted from Garibyan et al. [16]. In the case of EMS, an additional 189 mutations have been added (this work). Sequences in bold represent the three base pair sequence involving the largest hotspots.

Table 6G:C → A:T transitions resulting in nonsense mutations induced by UV in *lacI*^a.

Pyrimidine-pyrimidine			Purine-Pyrimidine		
Site	Sequence	Number of Occurrences	Site	Sequence	Number of Occurrences
A6	GAACC* AGGC	32	A9	CAAAC AGTC	6
A15	TGACC* AGGA	14	A31	CAAAC AGGA	5
A34	GGGCC*AGGC	8			
A16	TGACC AGAC	33	O9	GGCAC AACA	4
A19	TCACC AGCA	40	O10	ACAAC AACT	4
A21	ATGCC AGCC	10	O28	TCAAC AAAC	4
A24	ACTCC AGTC	21			
A5	TTATC AGAC	37	O11	GTCGC AAAT	2
A23	AATTC AGCC	145	O17	CGCGC AACG	3
A26	CGCTC AGAT	7	O21	CCAGC AAAT	3
A33	CTCTC AGGG	79	O29	CATGC AAAT	4
A35	CAATC AGCT	8	O34	GGGGC AAAC	2
O13	CGATC AACT	7	O35	GCTGC AACT	3
O24	CAATC AAAT	75			
O27	TTTTC AACA	106			
Total:		622	Total:		40

^a Data from Miller [20].**Table 7**

Triplets 5' – mutated base – 3'.

C → T	<i>rpoB</i> EMS Occurrences	<i>lacI</i> missense EMS ^a Occurrences
TCT/AGA	1, 10, 2	99, 55
TCC/GGA	46, 76	40
TCG/CGA	13	
TCA/TGA		15
CCT/AGG	11	
CCC/GGG		123
CCG/CGG	2,0,0	167, 0*
CCA/TGG		14
GCT/AGC		
GCC/GGC	0	59, 53
GCA/TGC	26	70, 24
GCG/CC		70, 83, 1
ACT/AGT		80
ACC/GGT	0	43
ACA/TGT		16, 16, 18
ACG/CGT	27	50, 8, 1

^a Data from [13].

second tier nearest neighbor preference is not straight forward in the case of NQO. However, we can see again an example of third tier preference, since the three TCT sites at positions 1535, 1565, and 1721 (see also Table 5) are represented by 8, 40, and 0 occurrences, respectively. Clearly, even though 1565 and 1721 have the same nearest neighbors, they are nearly two orders of magnitude different in their mutability by NQO. Our previous work in *lacI* showed that there is an almost 8-fold variation among some sites that are part of the same triplet [11].

Base Analogs, 2-aminopurine (2AP), zebularine (ZEB) and 5-azacytidine (5AZ). From Fig. 6 (Top Panel), we can see that for 2AP (data from our studies [16,30,49]) in *rpoB*, there is again a wide distribution of frequencies among the transition sites, the favored type of base change stimulated by 2AP. Among the G:C -> A:T sites, the most favored one is at position 1576, with 92 occurrences, whereas several sites have between zero and three occurrences (1546, 1565, 1600, 1714, 1721). 1576 is the only site where the C -> T change is part of a GCA triplet. Data from our *lacI* studies shows that this is certainly a favored triplet, but that there is still a 4-fold variation amongst those sites (third tier, [11]).

ZEB (Fig. 6, Middle Panel) has a dramatic hotspot at position 1600, with 102 occurrences at this site (out of 156 sequenced mutations). This hotspot site is part of a CGG/CCG triplet, yet, there are two other sites

that are also part of a CGG/CCG triplet, one of which (1609) is only 8 base pairs away. The other is at position 1708. These two other sites have only two and four occurrences, respectively. This is an extreme example of a third tier hotspot. A secondary ZEB hotspot, at position 1691, has 22 occurrences, and is the only site that is part of a CCT/AGG triplet.

5AZ has a unique specificity, generating only G:C -> C:G transversions [14,16,50,51]. Two prominent hotspots are evident, at 1576, and at 1716. The latter is part of a TCA/TGA triplet. The site at 1537 is also part of a TCA/TGA triplet, but is not a hotspot, with only 4 occurrences, compared to 43 at 1716. Again, we can see an example of a third tier hotspot.

Spontaneous mutations in different backgrounds. Fig. 7 depicts sets of mutations that we have never examined with this true distance metric. The top panel shows the 253 spontaneous mutations that are in the main cluster in *rpoB*. 40 additional mutations at a single site in a different region of *rpoB* (position 443) are not shown here, as this one site is poorly represented in most mutational spectra. One can see that mutations are distributed amongst three sub-regions of the 205 bp segment shown here. The first sub-region extends from 1522 to 1552, the second from 1565 to 1608, and the third from 1691 to 1721. There are many factors that can influence spontaneous mutation frequencies, such as replication errors, endogenous DNA damage, pool imbalances, and polymerase preferences. Although it is difficult to definitively attribute specific spontaneous mutations to a specific cause, the comparison of spectra provides clues. Thus, the two A:T -> G:C hotspots at 1547 and 1534 mimic the two most prominent hotspots seen in the mismatch repair deficient spectrum, suggesting that these two spontaneous hotspots arise from DNA polymerase errors. The A:T -> C:G hotspot at 1714 might arise from unrepaired oxidative damage of dGTP precursors, as this is the base change occurring in MutT-deficient strains (e.g. [14,16]). Also, the transversions at 1576 may result from a small contribution to replication errors by Pol IV and/or Pol V ([19]; see below).

The second panel maps strain EW100 [19], that carries a plasmid that overproduces the SOS polymerase Pol IV. Here the transversions at 1576 are the most prominent difference from the wild-type spectrum, and thus these can be clearly attributed to Pol IV. The positions of the mutations occurring in a MMR-deficient strain (third panel) are striking, in that they are largely confined to a 15 bp region, despite the availability of many sites (Table 5) for either the A:T -> G:C or the G:C -> A:T transitions that are the most frequent base substitutions in

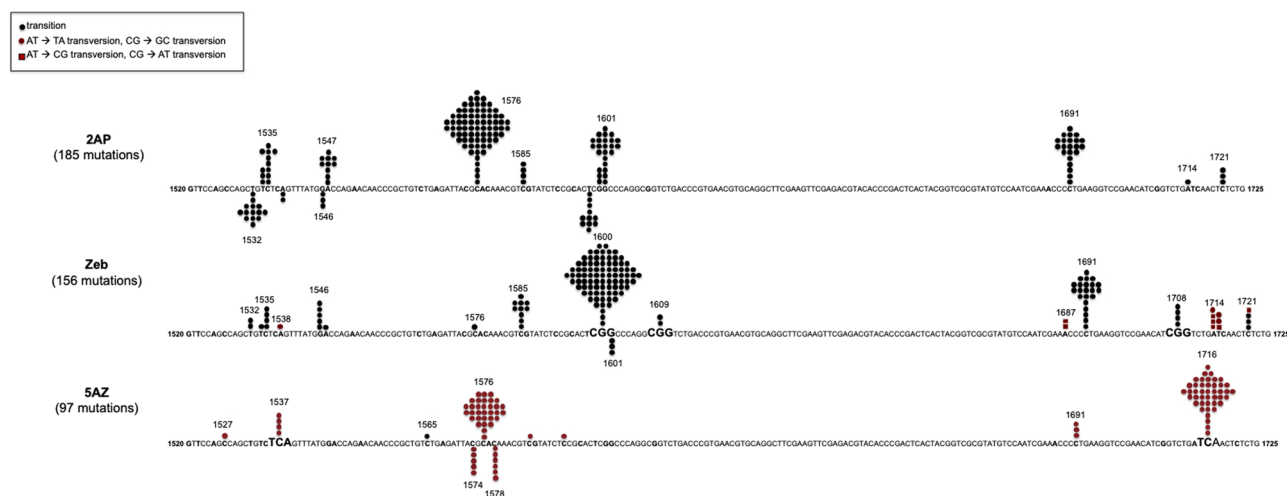


Fig. 6. Distribution of mutations in *E. coli rpoB*. For 2AP, the 185 mutations consist of 95 derived in CC107 [16,49], and 90 from strain BW25113 [30]. The ZEB-derived mutations are in CC107 [24], and the 5AZ mutations consist of 53 mutations from CC107 ([16]; this work) and 44 mutations from BW25113 [30].

MMR- strains. Foster and coworkers [18] looked at mutations occurring across the genome in a MMR- background and found that the A:T - > G:C transitions were most prominent at sites where the A is followed by a C, which we can depict as AC/GT, where AC represents the 5' - > 3' sequence on one strand, and the GT as the 5' - > 3' sequence on the other strand. This is indeed the case for the two prominent sites at 1547 and 1534, that have 137 and 58 occurrences, respectively. However, there are two other such AC/GT sites, at positions 1577 and 1703 with 11, and 0 occurrences, respectively. Within the 15 bp region are two A:T - > G:C sites with 16 and 11 occurrences (even though they are not part of a AC/GT). Also, the only GC - > A:T site with more than 3 occurrences in this set is at 1546, within the 15 bp region, with 25 occurrences. Foster and coworkers [18] reported regional bias in the distribution of MMR- mutations across the genome, although these regions are much larger than the 15 bp region here (from 1532 to

1547). Something in this small region make it particularly prone to DNA replication errors that are revealed in a MMR- background. (Schaaper and et al. [22] detected a preference based on the next added base that is valid for the preferences for the A:T - > G:C changes within the 15 bp region described here, but does not explain the finding of only a single occurrence at 1598. Also, it does not pertain to the G:C - > A:T changes, e.g. 1546).

5. Conclusions: future directions

What determines which mutational sites become hotspots, among a set of sites with identical nearest neighbor sequences? Here, we have analyzed mutational spectra to document how extensive this phenomenon is. In some cases, such as ZEB and CPT induced mutations, the sites with identical nearest neighbors but vastly different mutation

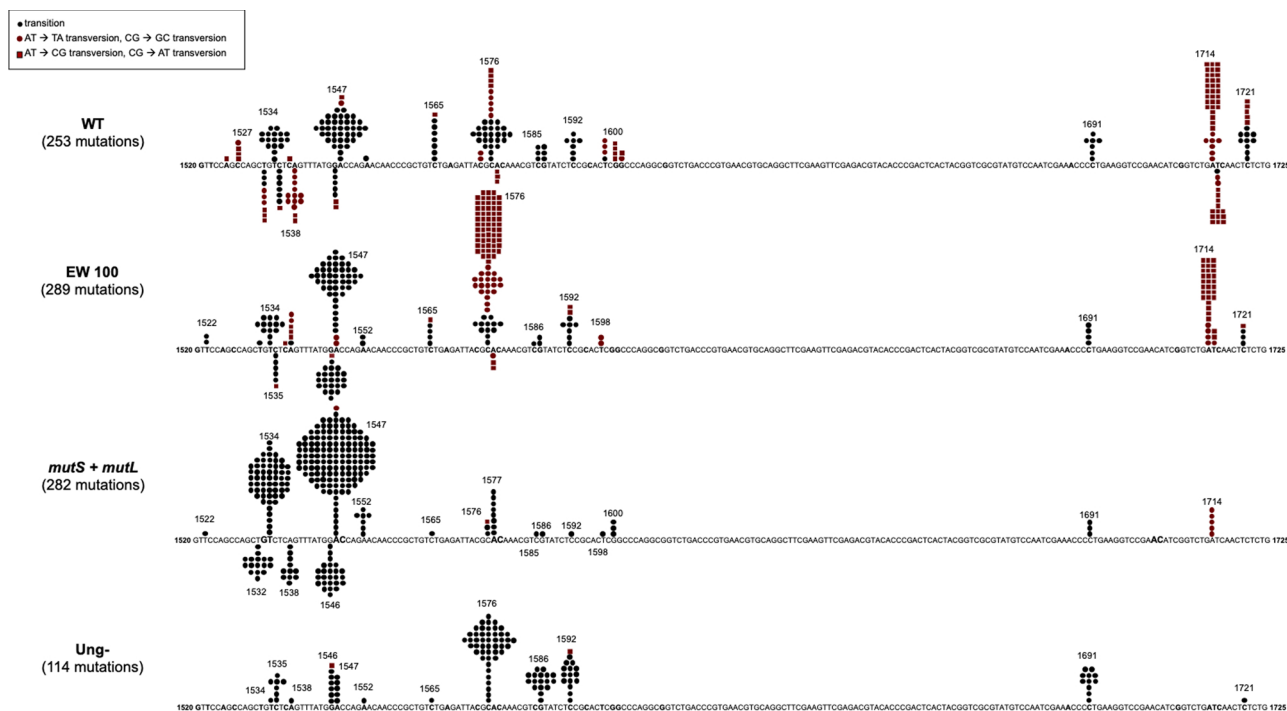


Fig. 7. Distribution of mutations in *E. coli rpoB* in wild-type and repair-deficient backgrounds, replotted. Top panel; mutations in the CC107 WT background [19]. Second panel; mutations in a strain (EW100) over-expressing Pol IV [19]. Third panel, mutations from several mismatch repair-deficient strains, CC107mutS [16], EM90 (CC107mutS lacking the F' plasmid), and BW25113mutL (this work). Bottom panel, mutations from BW25113ung (this work).

frequencies are fairly close together (8 and 6 bp, respectively). However, in the case of MMR-deficient cells, all the most mutable sites are clustered together within a 15 bp region. Several types of future experiments would be valuable. One should expand the number of different systems to examine mutational hotspots. Also, one can use different site directed change systems, such as bacterial recombineering [52] to alter specific bases near hotspots. Perhaps the most important experiment would be to analyze the 3D structure of short segments of DNA from well characterized systems, such as *rpoB*. Thus, comparing the structure of a 20 bp segment including the 1532–1547 sequence, a similar sized segment including the 1576 site, and one including the 1721 site (UV hotspot) would be of great interest. The 1576 site is of particular interest, as it is a mutational hotspot for several mutagens and for overproduction of Pol IV. In fact, the study of the 3D structure of short (20–30 bp) DNA duplex segments along a larger stretch of DNA in general would be of great value, and might reveal new insights.

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References

- [1] S. Benzer, E. Freese, Induction of specific mutations with 5-bromouracil, *Proc. Natl. Acad. Sci. U. S. A.* 44 (1958) 112–119.
- [2] S. Benzer, On the topography of the genetic fine structure, *Proc. Natl. Acad. Sci. U. S. A.* (1961) 403–415.
- [3] F.H. Crick, L. Barnett, S. Brenner, R.J. Watts-Tobin, General nature of the genetic code for proteins, *Nature* 192 (1961) 1227–1232.
- [4] E. Freese, The specific mutagenic effect of base analogs on phage T4, *J. Mol. Biol.* 1 (1959) 87–105.
- [5] J.W. Drake, R.H. Baltz, The biochemistry of mutagenesis, *Annu. Rev. Biochem.* 45 (1976) 1–37.
- [6] J.H. Miller, Perspective on mutagenesis and repair: the standard model and alternate modes of mutagenesis, *Crit. Rev. Biochem. Mol. Biol.* 40 (2005) 155–179.
- [7] E.C. Friedberg, G.C. Walker, W. Seide, R.D. Wood, R.A. Schultz, T. Ellenberger, *DNA Repair and Mutagenesis*, American Society of Microbiology, Washington, DC, 2006.
- [8] C. Yanofsky, J. Ito, V. Horn, Amino acid replacements and the genetic code, *Cold Spring Harbor Symp. Quant. Biol.* 31 (1966) 151–162.
- [9] L. Prakash, F. Sherman, Mutagenic specificity: reversion of iso-1-cytochrome c mutants of yeast, *J. Mol. Biol.* 79 (1973) 65–82.
- [10] B.N. Ames, F.D. Lee, W.E. Durston, An improved bacterial test system for the detection and classification of mutagens and carcinogens, *Proc. Natl. Acad. Sci. U. S. A.* 70 (1973) 782–786.
- [11] C. Coulondre, J.H. Miller, Genetic studies of the lac repressor IV. Mutagenic specificity in the lacI gene of *Escherichia coli*, *J. Mol. Biol.* 117 (1977) 577–60.
- [12] P.J. Farabaugh, U. Schmeissner, M. Hofer, J.H. Miller, Genetic studies of the lac repressor. VII. On the molecular nature of spontaneous hotspots in the lacI gene of *Escherichia coli*, *J. Mol. Biol.* 126 (1978) 847–863.
- [13] M. Pienkowska, B.W. Glickman, A. Ferreira, M. Anderson, M. Zielenska, Large-scale mutational analysis of EMS-induced mutation in the lacI gene of *Escherichia coli*, *Mutat. Res.* 288 (1993) 123–131.
- [14] C. Cupples, J.H. Miller, A set of lacZ mutations in *Escherichia coli* that allow rapid detection of each of the six base substitutions, *Proc. Natl. Acad. Sci. U. S. A.* 86 (1989) 5345–5349.
- [15] C.G. Cupples, M. Cabrera, C. Cruz, J.H. Miller, A set of lacZ mutations in *Escherichia coli* that allow rapid detection of specific frameshift mutations, *Genetics* 125 (1990) 275–280.
- [16] L. Garibyan, T. Huang, T.M. Kim, E. Wolff, A. Nguyen, T. Nguyen, A. Diep, K. Hu, A. Iverson, H. Yang, J.H. Miller, Use of the rpoB gene to determine the specificity of base substitution mutations on the *Escherichia coli* chromosome, *DNA Repair (Amst.)* 2 (2003) 593–608.
- [17] P.L. Foster, A.J. Hanson, H. Lee, E.M. Popodi, H. Tang, On the mutational topology of the bacterial genome, *G3 Bethesda (Bethesda)* 3 (2013) 399–407.
- [18] P.L. Foster, B.A. Niccum, E. Popodi, J.P. Townes, H. Lee, W. Mohammedsmail, H. Tang, Determinants of base pair substitution patterns revealed by whole-genome sequencing of DNA mismatch repair defective *Escherichia coli*, *Genetics* 209 (2018) 1029–1042.
- [19] E. Wolff, M. Kim, K. Hu, H. Yang, J.H. Miller, Polymerases leave fingerprints: analysis of the mutational spectrum in *Escherichia coli* rpoB to assess the role of polymerase IV in spontaneous mutation, *J. Bacteriol.* 186 (2004) 2900–2905.
- [20] J.H. Miller, Mutagenic specificity of ultraviolet light, *J. Mol. Biol.* 182 (1985) 45–68.
- [21] R.M. Schaaper, R.L. Dunn, Spectra of spontaneous mutations in *Escherichia coli* strains defective in mismatch correction: the nature of in vivo DNA replication errors, *Proc. Natl. Acad. Sci. U. S. A.* 84 (1987) 6220–6224.
- [22] D. Ahluwalia, R.M. Schaaper, Hypermutability and error catastrophe due to defects in ribonucleotide reductase, *Proc. Natl. Acad. Sci. U. S. A.* 110 (2013) 18596–18601.
- [23] P.M. Leong, H.C. Hsia, J.H. Miller, Analysis of spontaneous base substitutions generated in mismatch-repair deficient strains of *Escherichia coli*, *J. Bacteriol.* 168 (1986) 412–416.
- [24] G. Lee, E. Wolff, J.H. Miller, Mutagenicity of the cytidine analog zebularine in *Escherichia coli*, *DNA Repair (Amst.)* 3 (2004) 155–161.
- [25] D.H. Phillips, Mutational spectra and mutational signatures: insights into cancer aetiology and mechanisms of DNA damage and repair, *DNA Repair (Amst.)* 71 (2018) 6–11.
- [26] S.W. Piraino, S.J. Furney, Identification of coding and non-coding mutational hotspots in cancer genomes, *BMC Genomics* 18 (2017), <https://doi.org/10.1186/s12864-016-3420-9>.
- [27] S.M. Hecht, *Bleomycin: Chemical, Biochemical and Biological Aspects*, Springer, New York.
- [28] W. Nakahara, F. Fukuoka, T. Sugimura, Carcinogenic action of 4-nitroquinoline-N-oxide, *Gann* 48 (1957) 129–137.
- [29] J.H. Miller, P. Funchain, W. Clendenin, T. Huang, A. Nguyen, E. Wolff, A. Yeung, J. Chiang, L. Garibyan, M.M. Slupsky, H. Yang, *Escherichia coli* strains (ndk) lacking nucleoside diphosphate kinase are powerful mutators for base substitutions and frameshifts in mismatch repair deficient strains, *Genetics* (2002) 5–13.
- [30] J. Ang, L.J. Song, S. D'Souza, I.L. Hong, R. Lohar, M. Yung, J.H. Miller, Mutagen synergy: hypermutability generated by specific pairs of base analogs, *J. Bacteriol.* 198 (2016) 2776–2783.
- [31] M.M. Zambrano, D.A. Siegele, M. Almiron, A. Tormo, R. Kolter, Microbial competition: *Escherichia coli* mutants that take over stationary phase cultures, *Science* 259 (1993) 1757–1760.
- [32] K. Yamanaka, I.G. Minko, S.E. Finkel, M.F. Goodman, R.S. Lloyd, Role of high-fidelity *Escherichia coli* DNA polymerase I in replication bypass of a deoxyadenosine DNA-peptide cross-link, *J. Bacteriol.* 193 (2011) 3815–3822.
- [33] T. Baba, T. Ara, M. Hasegawa, Y. Takai, Y. Okumura, M. Baba, K.A. Datsenko, M. Tomita, B.L. Wanner, H. Mori, Construction of *Escherichia coli* K-12 in-frame single-gene knockout mutants: the Keio collection, *Mol. Sys. Bio.* (2006) 2006.0008.
- [34] K.A. Datsenko, B.L. Wanner, One-step inactivation of chromosomal genes in *Escherichia coli* K-12 using PCR products, *Proc. Natl. Acad. Sci. U. S. A.* (2000) 6640–6645.
- [35] J.H. Miller, *A Short Course in Bacterial Genetics: a Laboratory Manual and Handbook for Escherichia coli and Related Bacteria*, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y., 1992.
- [36] J.H. Miller, *Experiments in Molecular Genetics*, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y., 1972.
- [37] W.J. Dixon, F.J. Massey Jr., *Introduction to Statistical Analysis*, McGraw-Hill, New York, N.Y., 1969.
- [38] E. Curti, J.P. McDonald, S. Mead, R. Woodgate, DNA polymerase switching: effects on spontaneous mutagenesis in *Escherichia coli*, *Molec. Microbiol.* 71 (2009) 315–331.
- [39] P. Daubersies, S. Galiegue-Zoutina, N. Koffel-Schwartz, R.P.P. Fuchs, M.H. Loucheux-Lefebvre, B. Bailleul, Mutation spectra of the two guanine adducts of the carcinogen 4-nitroquinoline 1-oxide in *Escherichia coli*. Influence of the neighboring base sequence on mutagenesis, *Carcinogenesis* 13 (1992) 349–354.
- [40] B. Bailleul, P. Daubersies, S. Galiegue-Zoutina, M.-H. Loucheux-Lefebvre, Molecular basis of 4-nitroquinoline-1-oxide carcinogenesis, *Jpn. J. Cancer Res.* 80 (1989) 691–697.
- [41] D.A. Kreutzer, J.M. Essigman, Oxidized, deaminated cytosines are a source of C- & T transitions in vivo, *Proc. Natl. Acad. Sci. U. S. A.* 95 (1998) 3578–3582.
- [42] M.L. Michaels, J. Tchou, A.P. Grollman, J.H. Miller, A repair system for 8-oxo-7,8-dihydrodeoxyguanine, *Biochemistry* 32 (1992) 10964–10968.
- [43] H. Hayakawa, K. Kamura, M. Sekiguchi, Role of uracil-DNA glycosylase in the repair of deaminated cytosine residues of DNA in *Escherichia coli*, *J. Biochem.* 84 (1978) 1155–1164.
- [44] D.J. Downes, M. Chonofsky, K. Tan, B.T. Pfannenstiel, S.L. Reck-Peterson, R.B. Todd, Characterization of the mutagenic spectrum of 4-Nitroquinoline 1-oxide (4-NQO) in *Aspergillus nidulans* by whole genome sequencing, *GenesGenomesGenetics* 4 (2014) 2483–2493.
- [45] C.H. Corzett, M.F. Goodman, S.E. Finkel, Competitive fitness during feast and famine: how SOS polymerases influence physiology and evolution in *Escherichia coli*, *Genetics* 194 (2013) 409–420.
- [46] L.F. Povirk, Bleomycin-induced mutagenesis in repackaged lambda phage: base substitution hotspots at the sequence C-G-C-C, *Mut. Res.* 180 (1987) 1–9.
- [47] C.K. Mirabelli, A. Ting, C.-H. Huang, S. Mong, S.T. Crooke, Bleomycin and talisomycin sequence specific strand scission of DNA: a mechanism of double strand cleavage, *Cancer Res.* 43 (1982) 2779–2785.
- [48] M. Takeshita, L.S. Kappan, A.P. Grollman, M. Eisenberg, I.H. Goldberg, Strand scission of deoxyribonucleic acid by neocarzinostatin, aurumycin and bleomycin: studies on base release and nucleotide sequence specificity, *Biochemistry* 20 (1981) 7599–7606.
- [49] S. D'Souza, J.E. Miller, J. Ahn, R. Subandi, D. Lozano, J. Ramirez, M. Goff, C. Davidian, J.H. Miller, The antibiotic trimethoprim displays strong mutagenic synergy with 2-aminopurine, *Antimicrob. Agents Chemother.* 63 (2019) e01577–18.
- [50] D.E. Levin, B.N. Ames, Classifying mutagens as to their specificity in causing the six possible transitions and transversions: a simple analysis using the Salmonella mutagenicity assay, *Environ. Mutagen.* 8 (1986) 9–28.
- [51] K.M. Doiron, J. Lavigne-Nicolas, C.G. Cupples, Effect of interaction between 5-azacytidine and DNA (cytosine-5) methyltransferase on C-to-G and C-to-T mutations in *Escherichia coli*, *Mut. Res.* 429 (1999) 37–44.
- [52] L.C. Thomason, J.A. Sawitzke, X. Li, N. Costantino, D.L. Court, Recombineering: genetic engineering in bacteria using homologous recombination, *Current Protocols Molec. Biol.* 1 (16) (2014) 1–39, <https://doi.org/10.1002/042727.mb0116s106>.